

CS & IT ENGINEERING



Operating System

Memory Management

Lecture – 5

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Recap of Previous Lecture

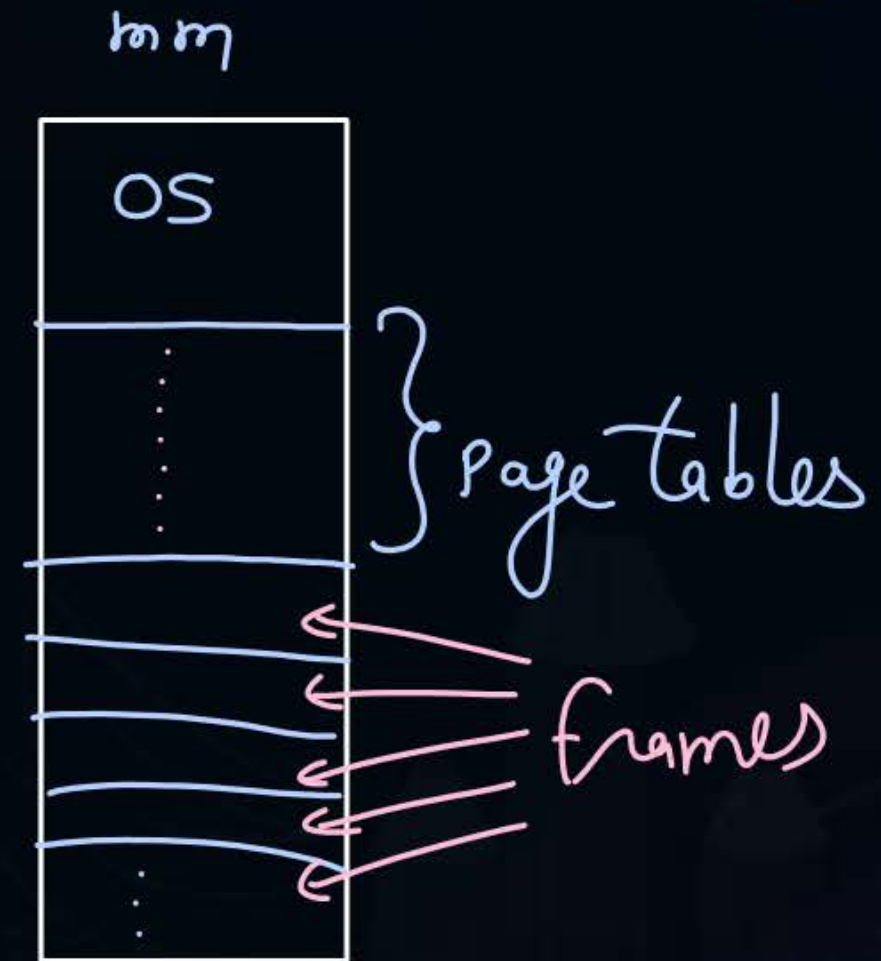


Topic

Paging

Topic

TLB



Topics to be Covered



Topic

TLB Mapping

Topic

Segmentation



Topic : Paging

How TLB Stores Entries?

CPU



P. Table P)



TLB
Contains

only
frame no. + extra bits



for every context
switch, TLB is
emptied.

Process id
+
frame no.
+
extra bits

all time TLB entries
present in TLB

TLB



4 entries

P.T.



6

15



16 P.T. entries
(16 pages in process)



Topic : Paging

TLB Mapping

1. Fully Associative
2. Direct
3. Set-Associative



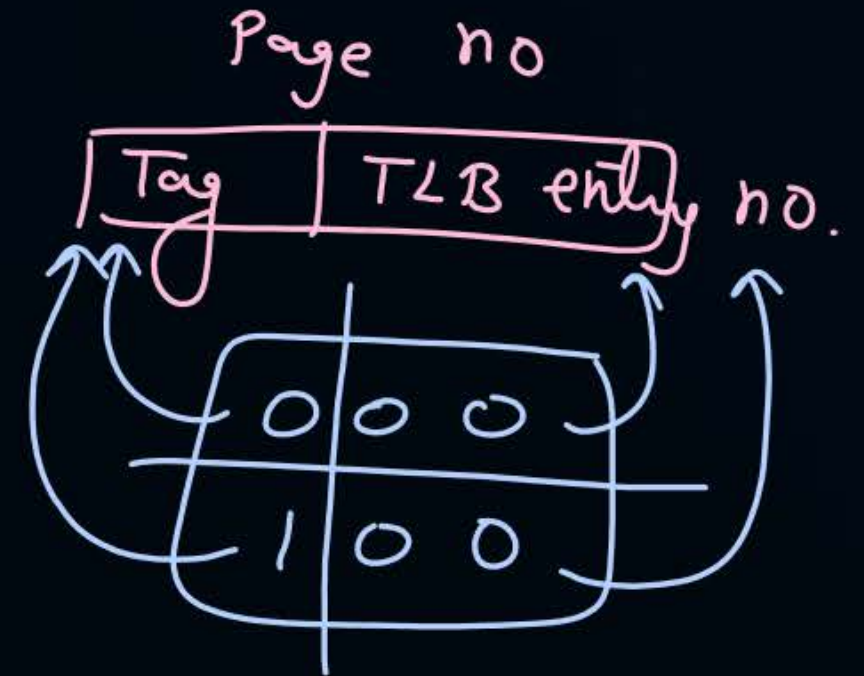
Topic : Paging

TLB Mapping: Direct

(1 bit) Tag	TLB	Pattern
00	☐	0, 4
01	☐	1, 5
10	☐	2, 6
11	☐	3, 7

	P.T.
000	
001	
010	
011	
100	
101	
110	
111	

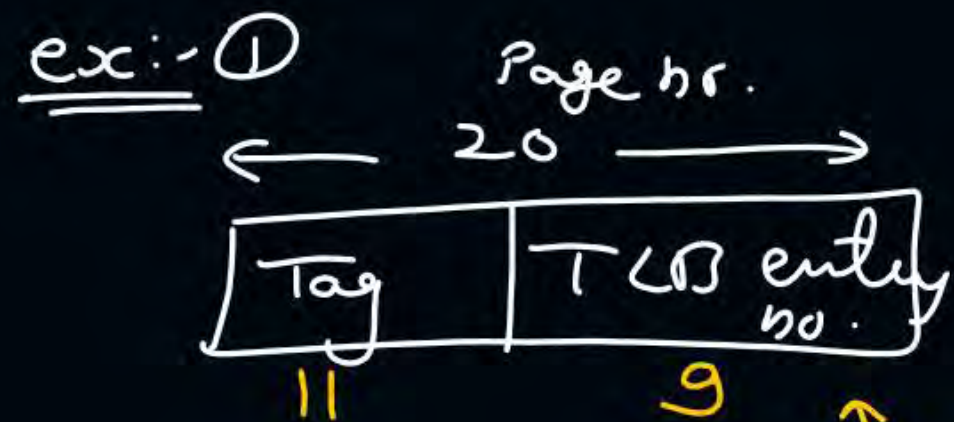
8 entries
P.T.



Page no.

Tag	TLB entry no.
-----	---------------

no. of bits for TLB entry no. = $\log_2 (\text{no. of entries stored in TLB})$



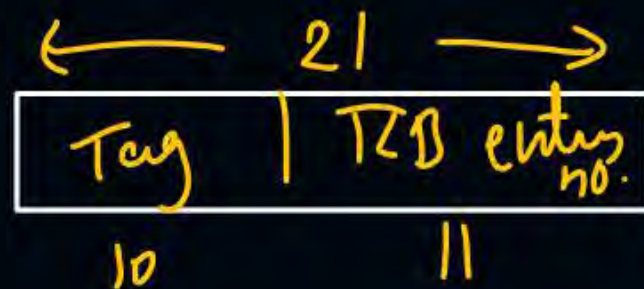
no. of entries in TLB = 2^9

Tag = 11 bits

② no. of entries in TLB = $2k = 2 * 2^{10} = 2^{11}$

Page no. = 21 bits

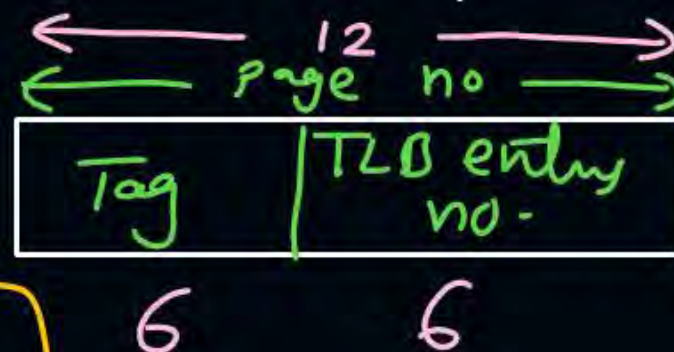
Tag = 10 bits



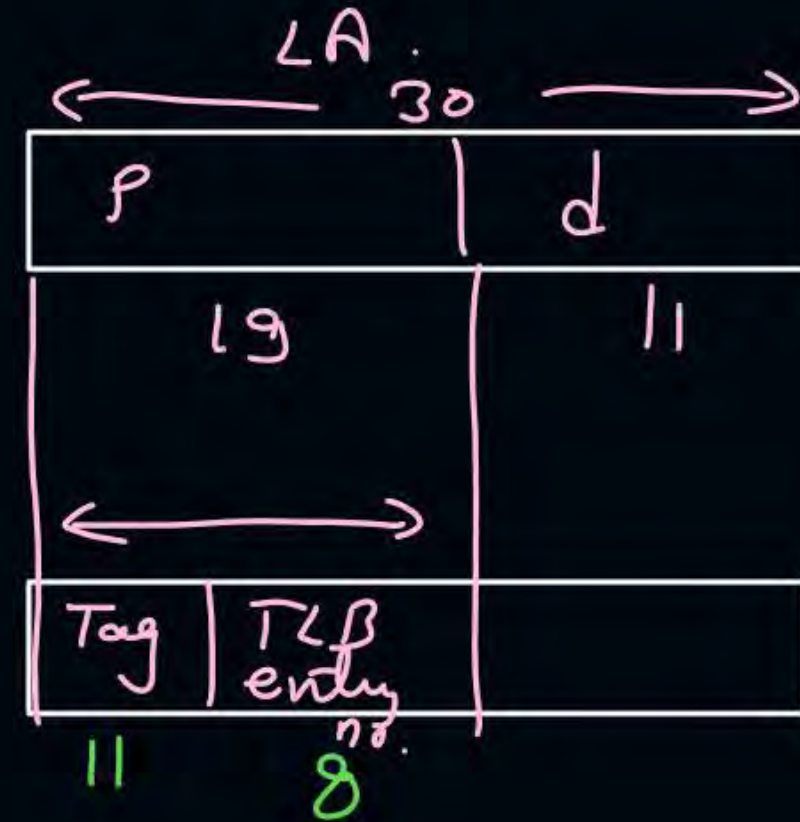
③ no. of pages in process = $4k = 2^{12}$

TLB can store = 64 entries = 2^6

Tag = 6 ?



Ques) $LA = 30 \text{ bits}$
 Page size = 2k bytes = 2^{11} B
 Direct mapping
 TLB can store $\Rightarrow 256 \text{ entries}$
 $\hookrightarrow 2^8$
 Tag = 11 ?



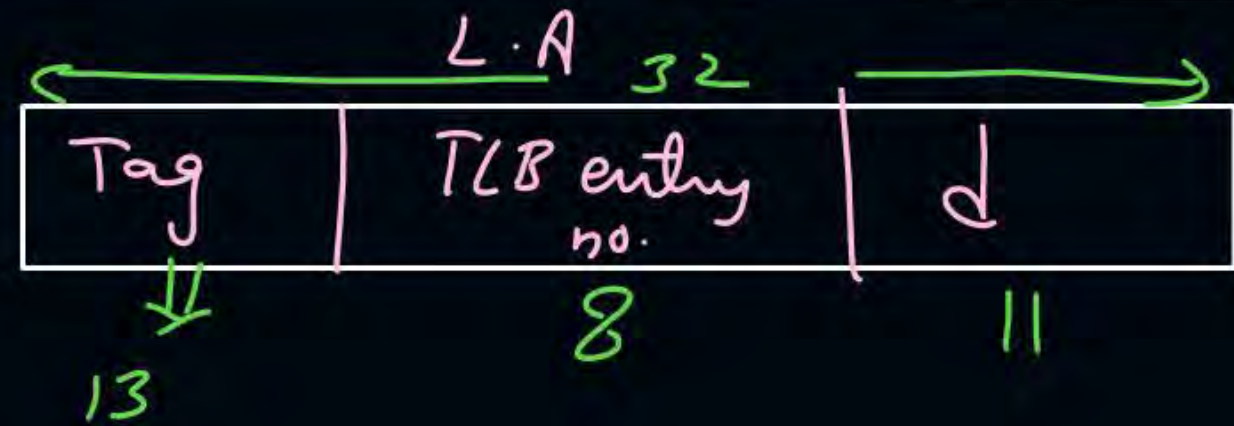
Ques) Direct mapped TLB, size = 1KB

TLB stores page table entries each of which is 4 bytes

L.A = 32 bits

Page size = 2KB = 2^{11} B

Tag = 13 bits?



$$\begin{aligned}\text{no. of entries in TLB} &= \frac{1\text{KB}}{4\text{B}} \\ &= \frac{2^{10}}{2^2} = 2^8\end{aligned}$$

Ques Direct mapped TLB can store 512 entries.

$\rightarrow 2^9$

$$\text{Page size} = 4\text{KB} = 2^{12}\text{B}$$

$$\text{Tag} = 12\text{ bits}$$

$$\text{L.A.S.} = \underline{8}\text{ GB?}$$

L.A.

Tag	TLB entry no.	d
-----	---------------	---

12

9

12

$\underbrace{\hspace{10em}}_{\text{L.A.} = 33\text{ bits}}$

$$\text{L.A.S.} = 2^{33}\text{ bytes} = 8\text{GB}$$



Topic : Paging

TLB Mapping: Set Associative

Page no.

Tag	TLB set no.
0	

TLB can store 8 entries
2-way set associative

00
01
10
11

4-way

0
1

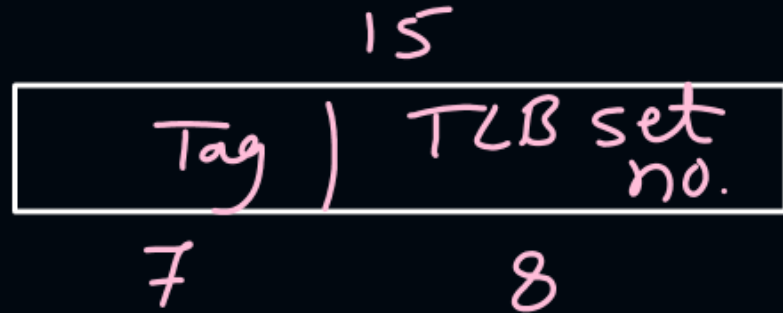
$$\text{no. of bits in TLB set no.} = \log_2 (\text{no. of sets in TLB})$$

$$\text{no. of sets in TLB} = \frac{\text{no. of entries in TLB}}{\text{associativity}}$$

ex:- TLB $\Rightarrow$ 512 entries
2-way set associative

Page no. = 15 bits

Tag = 7 bits



$$\text{no. of sets} = \frac{512}{2} = \frac{2^9}{2^1} = 2^8$$

Ques) LA = 31 bits

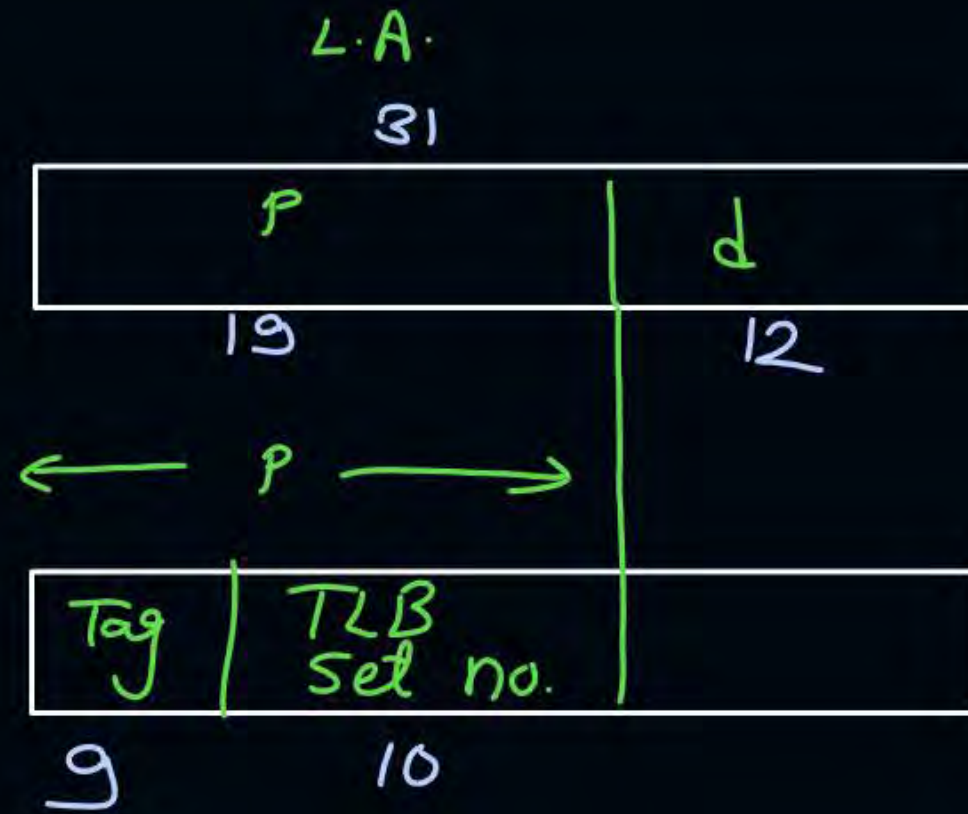
Page size = 4KB = 2^{12} B

TLB $\Rightarrow$ 4k entries

4-way set associative

Tag = 9 bits ?

$$\begin{aligned} \text{no. of sets in TLB} &= \frac{4K}{4} \\ &= 1K = 2^{10} \end{aligned}$$





Topic : Paging

TLB Mapping: Fully Associative → associative memory (content addressable memory)

L.A.



TLB

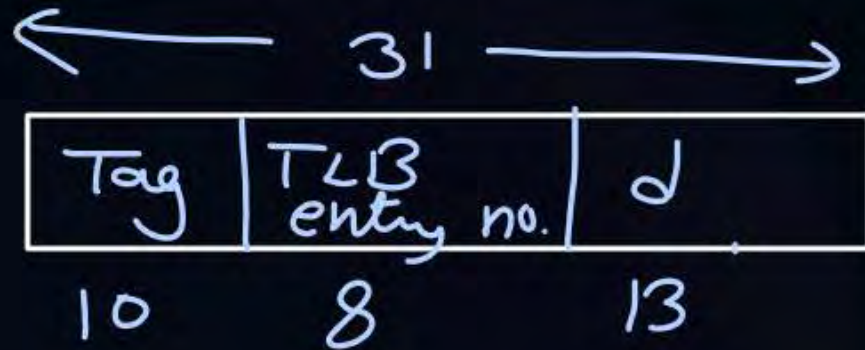
Page no.	Frame no. + extra bits
6	15 + ...
2	18 + ...
1	9 + ...
⋮	⋮

No any pattern to follow.



Topic : Question

#Q. A computer system implements a 31-bit virtual address, page size of 2^{13} (8) kilobytes, and a 256-entry translation look-aside buffer (TLB) organized as direct mapped. The minimum length of the TLB tag in bits is 10?





Topic : Question

#Q. A computer system implements a 44- bit virtual address, page size of 1 kilobytes, and a 16KB look-aside buffer (TLB) organized as direct mapped. Each page table entry is of 4bytes. The minimum length of the TLB tag in bits is 22?

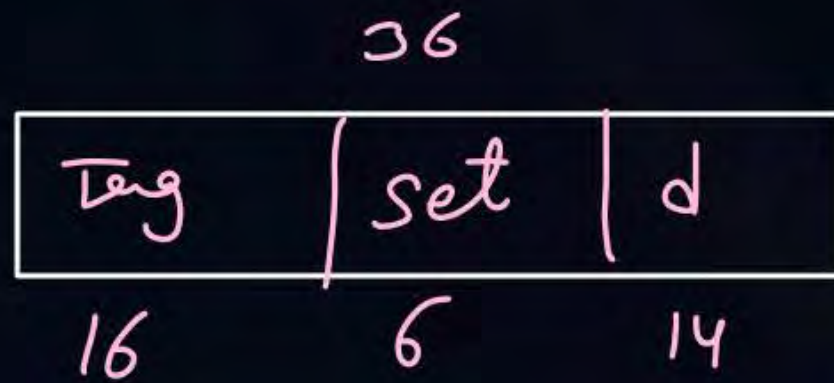
44		
tag	TLB entry no.	d
22	12	10

$$\text{no. of entries in TLB} = \frac{16 \text{ KB}}{4 \text{ B}} = 4 \text{ K} = 2^{12}$$



Topic : Question

#Q. A Computer system implements a 36-bit virtual address, page size of 16 Kbytes and a 256 – entry translation look-aside buffer (TLB) organized into 64 sets each having four ways. Assume that the TLB tag does not store any process id. The minimum length of the TLb tag in bits is 16.

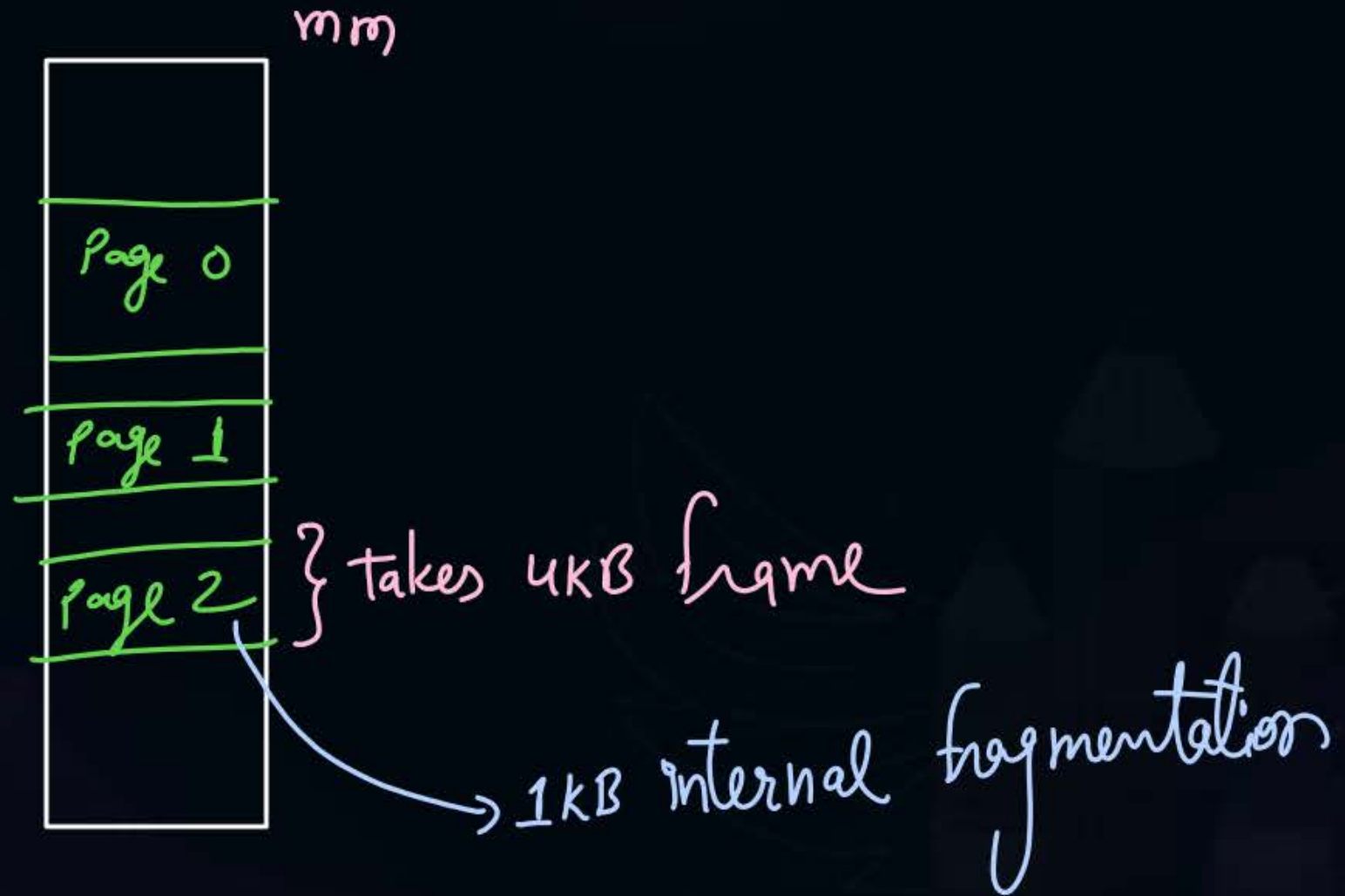
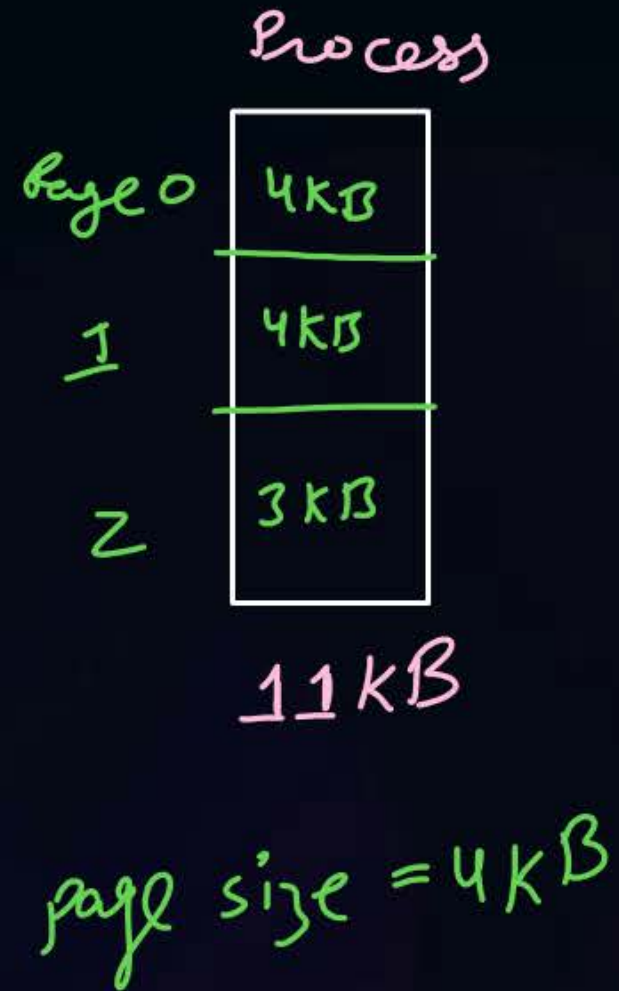


$$\frac{256}{4} = 64$$



Topic : Paging

Paging suffers from internal fragmentation



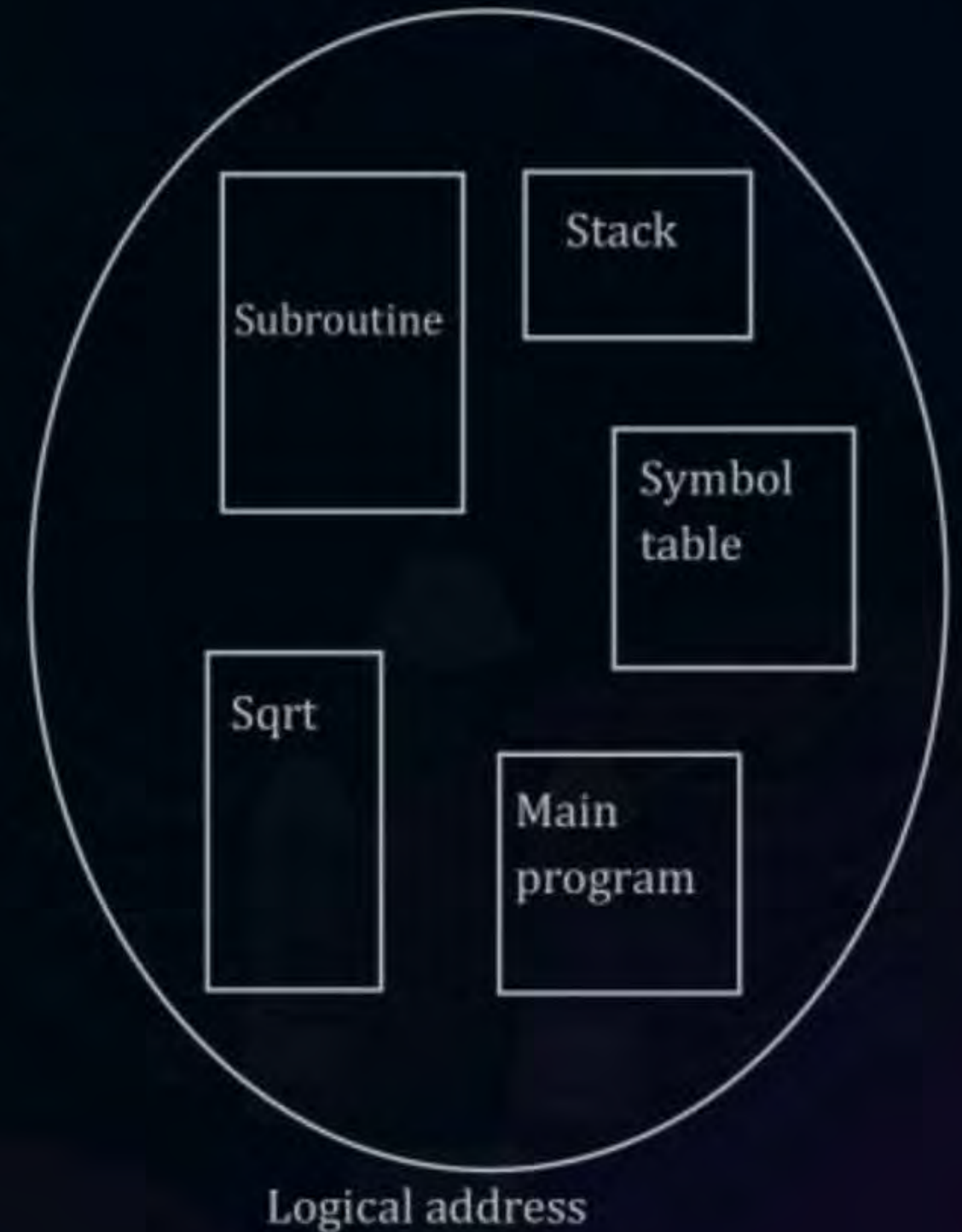


Topic : Segmentation

- Divide Process in logically related partitions (Segments) Segments are scattered in physical memory
- ~~Divide Process in logically related partitions (Segments)~~
- ~~Segments are scattered in physical memory~~

segments are of diff. sizes.

⇒ No any internal fragmentation.





2 mins Summary

Topic

TLB Mapping

Topic

Segmentation



Happy Learning

THANK - YOU